

# HDOC Day-36

Write a menu-driven C program to perform the following operations using separate functions:

1. Count characters in a string without using built-in length functions.
2. Print each character of a string on a new line.
3. Count vowels and consonants in a string.
4. Exit.

## Algorithm

**Step 1:** Start.

**Step 2:** Declare a character array `str[100]` and an integer variable `choice`.

**Step 3:** Read the string using `fgets()`.

**Step 4:** Display the menu:

1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

**Step 5:** Read the user's choice.

**Step 6:** Perform the selected operation using separate functions:

- **If choice = 1:**  
Call `countCharacters()` and count characters until `'\0'` is encountered. Display the count.
- **If choice = 2:**  
Call `printCharacters()` and print each character of the string on a separate line.
- **If choice = 3:**  
Call `countVowelsConsonants()`. Check each alphabet:
  - If it is a, e, i, o, u, increment vowel count.
  - Otherwise, increment consonant count.Display both counts.
- **If choice = 4:**  
Display "Exiting program" and terminate the program.

- **Otherwise:**  
Display "Invalid choice".

**Step 7:** Repeat the menu until the user selects **4 (Exit)**.

**Step 8:** Stop.

**Test Case Table:**

| Test Case | Input String | Choice | Expected Output                        |
|-----------|--------------|--------|----------------------------------------|
| 1         | Hello Kitty  | 1      | Number of characters= 12               |
| 2         | Hello Kitty  | 2      | H e l l o k i t t y in different lines |
| 3         | Hello Kitty  | 3      | Vowels=3 , consonants= 7               |
| 4         | Hello Kitty  | 4      | Exiting the program                    |
| 5         | Hello Kitty  | 5      | Invalid choice                         |

**Compilation and Execution:**

The following program has been compiled and executed successfully.

Test Case 1:

```
(kali@kali)-[~]
$ ./a.out
Enter a string: Hello kitty

===== MENU =====
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 1
Number of characters = 12
```

Test Case 2:

```
===== MENU =====
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 2
H
e
l
l
o

k
i
t
t
y
```

Test Case 3:

```
===== MENU =====
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 3
Number of vowels = 3
Number of consonants = 7
```

Test Case 4:

```
===== MENU =====
1. Count characters
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 4
Exiting program...
```

Test Case 5:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter a string: Hello kitty  
  
===== MENU =====  
1. Count characters  
2. Print each character on a new line  
3. Count vowels and consonants  
4. Exit  
Enter your choice: 5  
Invalid choice! Please try again.
```